REGULARITY OF A RING MAP IS PRESERVED BY BASE CHANGE ALONG A FINITE TYPE MAP

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Allowed standard prerequisites (editorial).

- Flatness is preserved by base change. Algebra, Lemma \ref{algebra-lemma-flat-base-change}, cited at proof lines 10420-10421. Gives the flatness half of regularity for the base changed map.
- Noetherianity of a finitely generated algebra over a field extension, and Noetherianity of the fibres. Algebra, Lemma \ref{algebra-lemma-Noetherian-field-extension}, cited at proof lines 10432-10434. Shows the fibres of the base changed map are Noetherian.
- Geometric regularity is preserved when extending the base field by a finitely generated extension. Algebra, Lemma \ref{algebra-lemma-geometrically-regular}, cited at proof lines 10435-10436. Gives geometric regularity of the base changed fibres.

Source reference. Original lemma 07C1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9361–9361. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00HI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 644–644. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 045I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5437–5437. Navigation pointer to a statement the

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Source reference. Original lemma 0381: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4220–4220. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2 (Regular maps and base change)**.** *Let $R \to \Lambda$ be a regular ring map. For any finite type ring map $R \to R'$ the base change $R' \to \Lambda \otimes_R R'$ is regular too.*

Proof. Flatness is preserved under any base change, see Algebra, Lemma 00HI. Consider a prime $\mathfrak{p}' \subset R'$ lying over $\mathfrak{p} \subset R$. The residue field extension $\kappa(\mathfrak{p}')/\kappa(\mathfrak{p})$ is finitely generated as R' is of finite type over R. Hence the fibre ring

$$(\Lambda \otimes_R R') \otimes_{R'} \kappa(\mathfrak{p}') = \Lambda \otimes_R \kappa(\mathfrak{p}) \otimes_{\kappa(\mathfrak{p})} \kappa(\mathfrak{p}')$$

is Noetherian by Algebra, Lemma 045I and the assumption on the fibre rings of $R \to \Lambda$. Geometric regularity of the fibres is preserved by Algebra, Lemma 0381. □